SAHIL KUMAR

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PROFESSIONAL SUMMARY

MCA (AI & ML) student at Sharda University with hands-on experience building AI-powered applications using Python, NLP, RAG pipelines, Large Language Models, and deep learning. Built an AI tutoring system with semantic search, a CNN-based facial emotion detection web app, a demand-forecasting and multi-tool business assistant, and a Power BI analytics dashboard. Solid foundation in Data Structures & Algorithms, Machine Learning, SQL, and data analysis. Looking for entry-level roles in Machine Learning, Data Science, Data Analytics, or Software Development.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (basic), C (Beginner), C++ (Beginner)
Data Structures & Algorithms: Arrays, Linked Lists, Stacks, Queues, Recursion, Searching & Sorting
Machine Learning & Deep Learning: CNNs, TensorFlow / Keras, TensorFlow.js, Demand Forecasting, Feature Engineering (lag & rolling features), Model Validation (MAPE, RMSE)
NLP & LLMs: Natural Language Processing, RAG Pipelines, Large Language Models, Prompt Engineering, Vector Embeddings
Data Analysis & Databases: NumPy, Pandas, Data Manipulation, EDA, SQL, Star Schema Design
Data Visualization / BI: Power BI, Dashboard Development, Data Reporting

PROJECTS

RetailIQ — Demand Forecasting & Multi-Tool Business Assistant | *Python · SQL · Machine Learning · RAG · LLM Agents*

- Designing an end-to-end retail intelligence application that forecasts demand at product-store-week level and answers natural-language business questions.
- Modelled retail data in a star schema (fact_sales with product, store, date, and promotion dimensions) and built a clean analytical dataset through EDA on seasonality, promotional lift, and price sensitivity.
- Developing forecasting models with lag features, rolling statistics, and promotional/calendar features, validated with forward-chained validation and reported using MAPE and RMSE.
- Building a planner-executor agent that selects between a SQL tool, a forecasting tool, and a RAG tool over policy documents to return grounded, source-cited answers.

Facial Emotion Detection Web App | *Python · TensorFlow/Keras · CNN · TensorFlow.js · FER2013* | GitHub

- Trained a CNN on the FER2013 dataset to classify facial expressions into 7 emotion categories, using Google Colab for model training.
- Diagnosed and fixed a model-collapse failure (loss stuck at ln(7)) using He-normal initialization, a reduced learning rate with gradient clipping, capped class weights, and label smoothing.
- Built a browser-based frontend in HTML/CSS/JavaScript using TensorFlow.js and face-api.js, running the model client-side with no backend required.

EduRAG Tutor — AI-Powered Tutoring System | *Python · NLP · RAG · LLMs · Vector Embeddings*

- Built an AI tutoring system enabling students to query educational materials in natural language.
- Designed a full RAG pipeline that retrieves relevant document chunks and generates context-aware responses using an LLM.
- Implemented vector-based semantic search across educational resources for accurate, low-hallucination answers.
- Applied prompt engineering techniques to improve response quality and relevance.

College Placement Analytics Dashboard | *Power BI · Data Visualization · Business Intelligence*

- Designed an interactive Power BI dashboard to track student placement trends and recruitment metrics.
- Visualized KPIs including placement rates, average salary packages, and department-wise outcomes.
- Used slicers, filters, cards, and charts to enable drill-down analysis for placement teams.

INTERNSHIP EXPERIENCE

Data Audit & Desktop Assessment Intern — Swachh Survekshan 2025–2026 60 Days · Jul 2026
Skillexpress

- Contributed to the Swachh Survekshan 2025-2026 initiative through data audit and desktop assessment activities supporting India's national urban sanitation ranking program.
- Applied analytical aptitude and attention to detail across evaluation and assessment processes to support cleaner, healthier, and more sustainable urban communities.
- Awarded a Certificate of Excellence in recognition of dedication and valuable contribution to the program (Certificate No. SESSU202674).

EDUCATION

Master of Computer Applications (MCA) — AI & Machine Learning 2025 – Present
Sharda University, Greater Noida, UP

Bachelor of Computer Applications (BCA) 2022 – 2025
Patliputra University, Patna, Bihar

Class XII (Intermediate) 2020 – 2022
St. Xavier's High School, Patna, Bihar